

3

Volatility Estimation in Energy Markets

Vince Kaminski, Grant Masson, and Ronnie Chahal of Enron Corp. Research Group

3.1 Introduction

Volatility is one of the critical concepts in option pricing and risk management. Organised exchanges use volatility of the underlying commodities to determine the required level of margins that futures traders are required to post with the clearing house. Volatility has been traditionally defined as the standard deviation of price returns and is routinely estimated from historical price series in countless spreadsheets all over the world. In spite of its popularity, the notion of volatility is often misunderstood, especially when one applies this tool to the energy markets. This concept has been defined and extensively researched in the context of financial instruments and one has to be very careful making generalisations. It is critical that in practical applications in the energy industry one makes an effort to capture the salient features of the energy markets. The points to make are:

- Volatility can be defined and estimated in a meaningful way only in the context of a specific stochastic process for the prices (price returns).
- The volatility definition and measure should capture the key features of energy markets, such as the seasonal dependence on the price level.

In the following we identify a number of practical problems of price process modelling in energy markets.

One implication of the assumption of the GBM process of equation (1.4) is the continuity of the line representing the sample trajectory of the price over time. The line could be drawn, in principle, without removing the pencil from the sheet of paper. The assumption of GBM as the process that describes the dynamics of the prices of financial instruments is an approximation of the behaviour observed in real markets and has to be treated as a stylised fact. As a matter of fact, there is growing evidence that the behaviour of market prices did not conform in many past time periods to this standard assumption of financial economics. One especially troubling observation is that the empirical frequency of the occurrence of extreme outcomes is larger than the probability implied by theoretical models. This issue will be revisited below.

The assumption of GBM strikes anyone with practical experience in energy commodity trading as an unrealistic description of the observed behaviour of energy prices. This has been recognised by a growing number of academics and practitioners who have devoted a lot of attention to developing more realistic models of energy commodity price behaviour, many of which are referred to in the following pages. What follows is a brief review of the most important issues that have been overlooked in early modelling efforts and have been fully addressed only in recent research. One should note that energy commodities are not created equal and many of the observations made below apply only to some subsets of their entire universe.

Investment assets vs. consumption goods. The most obvious observation is that energy commodities cannot be treated as purely financial assets, which are treated by owners as an investment. Energy commodities are inputs to production processes and/or consumption goods and this explains why many models based on a mechanical extension of the approach developed for financial markets may break down in the case of energy related contingent claims. For example, the GBM assumption does not allow negative prices. This assumption may be violated in practice often enough to require attention, especially in the case of electricity. In some cases, prices of electricity bid into a power pool may drop to zero if some generators want to guarantee that their plants are dispatched for contiguous blocks of time, longer than a single time slot for which separate bids are accepted. In some cases, the price may become negative, as power plants have to get rid of excess output and have no option to store electricity. In other words, an assumption of free disposal, customarily made in theoretical models, does not hold. This problem has been addressed by some recently published papers (see Routledge, Seppi, Spatt (1999)).

Prices of energy commodities display seasonality. By this we mean recurring regularities in price levels and/or price volatility observed over time. Seasonality may correspond to the time of the year (winter versus summer versus shoulder month), or may be observed in intramonth, intraweek, and in some markets (like power) intraday prices. Seasonality results primarily from regular demand fluctuations, driven in most cases, by recurring weather related factors. Fluctuations in demand interact with the supply side factors: increased demand can be satisfied only from more expensive sources or by using more expensive production units. In many cases, increased demand resulting from weather related factors might reach the levels at which supply becomes constrained by the capacity of the existing transportation or the transmission grid. In many markets, seasonality may change over time due to the changes in economic conditions and technology. For example, many natural gas marketers expect a change in seasonal price patterns in the US natural gas markets starting in year 2000, due to increases in the gas-fired generation capacity. It is expected that in addition to the winter peak, one will observe a more pronounced July/August peak, related to air conditioning load.

Recognition of the existing and possibly changing patterns of seasonality creates a need for forward-looking modelling. The information about future seasonality is often derived in formal models from the futures/forward prices that summarise all the

information available to the market about future demand and supply patterns. Some recent papers offer ingenious methods of calibrating prices to forward price curves (see for example Clewlow and Strickland (1999)).

Commodity prices often display jump behaviour. Jump behaviour, or ‘gapping’, is driven in many cases by fluctuations in demand and low elasticity of supply, reflecting rigidities in the transportation and transmission system and limited inventories. We saw in section 2.7 a model to incorporate this effect.

Prices gravitate to the cost of production. The assumption of GBM allows prices to wander off to unrealistic levels. The same approach, used in the modelling of two related commodities, like natural gas and power, or peak and off-peak electricity prices, may produce unrealistic spreads between them. The departures from the cost of production, or “normal” price spreads is possible in the short run under abnormal market conditions, but in the long-term, the supply will be adjusted and the prices will move to the level dictated by the cost of production. This adjustment can be captured via mean reversion which was introduced in chapter 2. One could argue that the use of a mean-reversion process represents another case of looking for the car keys under the street light, even if they were lost somewhere else. Vasicek (1977) first used a mean reverting model for modelling interest rate dynamics and subsequently the model was widely adapted. In the case of energy commodities, a pure mean reversion model may not perform well. First of all, the speed of mean reversion may be different below and above the long-term level. Secondly, in many markets, especially in the case of electricity, one can expect more departures to the upside, than to the downside. Thirdly, a price spike in one direction is frequently neutralized by a spike of similar magnitude and opposite sign, occurring shortly after the initial spike. The mean reversion process generally produces a re-adjustment that is less abrupt.

Prices of energy commodities behave differently during different periods of their lives. This is especially true of forward prices. According to the so-called Samuelson’s hypothesis, forward price volatility increases as they get closer to their maturity. This can be explained by the fact that more information becomes available as the forward contract gets closer to delivery period, and this results in more trading, which in turn produces more volatility. The authors believe that GBM may represent a reasonable approximation to the reality of forward markets. Once a forward contract reaches maturity and we enter the delivery period, the behaviour of prices becomes more erratic and subject to frequent jumps. This suggests that one can model price behaviour using more traditional apparatus like GBM or mean reversion during the forward stages of their life, switching to more complex processes to describe the dynamics of the spot price during the delivery month.

3.2 Estimating Volatility

In this section we look at issues concerning the estimation of volatility including both historical and implied measures.

3.2.1 Estimation of Volatility From Historical Data

The first step in the determination of the correct level of volatility is the examination of the historical price data. In the case that the underlying spot price process is assumed to be GBM, volatility can be estimated from the historical price returns. The process can be broken into several steps that can be easily carried out in a spreadsheet.

Step 1. Calculate logarithmic price returns.

This can be accomplished by forming the price ratios S_t/S_{t-1} and taking the natural logarithms of these ratios. Price returns are typically calculated as $r = S_t/S_{t-1} - 1$. The logic of the approach described above is that for relatively small x , $\ln(1 + x) \cong x$. Taking the natural log of S_t/S_{t-1} is equivalent to taking the natural log of $1 + r$, and this in turn is roughly equal to r .

The use of natural log returns has also some other additional advantages. If one wants to calculate a log return over a longer time period, say from t to $t + n$, corresponding to the ratio S_{t+n}/S_t , one can convert this into $(S_{t+n}/S_{t+n-1}) \times (S_{t+n-1}/S_{t+n-2}) \times \dots \times (S_{t+1}/S_t)$. Given that the log of a product is equal to the sum of the logs, one can easily show that a log return over the longer time period can be calculated as the sum of log returns for the sub periods.

Step 2. Calculate standard deviation of the logarithmic price returns.

Step 3. Annualise the standard deviation by multiplying it by the correct factor.

As a first approximation the annualisation factor depends on the price data frequency. In the case that the data is monthly, the factor is $\sqrt{12}$; for weekly data it is $\sqrt{52}$. For the daily data available for each calendar day one has to use $\sqrt{365}$. If the information is available for trading days only, one should use the relevant number that may vary from jurisdiction to jurisdiction. The standard usage is $\sqrt{250}$.

The logic of annualisation comes from the assumptions regarding additivity of variance. It is implicitly assumed in this approach that each period return is drawn from a certain probability distribution and we are estimating the variance of this unknown distribution from the time series data for the market closing prices, ignoring the information about price behaviour within the period (intraday, intramonth, etc.). If the price returns for each period are iid (independent, and identically distributed) random variables, the variance of the sum of n random variables, $x_1, x_2, \dots, x_n$, is equal to the sum of the variances (if price returns are uncorrelated):

$$\sigma_{x_1+x_2+\dots+x_n}^2 = \sigma_{x_1}^2 + \sigma_{x_2}^2 + \dots + \sigma_{x_n}^2 \quad (3.1)$$

Given that all the variances are assumed to be equal, the variance of the sum is equal to the common variance of the underlying variables multiplied by n . Therefore, the standard deviation is equal to the common standard deviation multiplied by the square root of n .

Before we address the annualisation problem in more depth we offer a few warnings to try to avoid common pitfalls. Firstly, energy price series tend to be of low quality in

many cases. It is therefore necessary to screen the price history for data errors that may significantly distort the estimates of volatility. Secondly, in many cases the price data is obtained from the organised futures exchanges by stitching together the numbers representing closing prices for the first available contract. One should keep in mind that in calculating price returns, one should discard the observation corresponding to the contract rollover date (the day following the expiration of one contract, when we switch to the price data representing the new prompt contract). The logic of this approach is that one can, hopefully, mitigate the seasonality problem in the price data. Assuming that the prices corresponding to the same contract month are characterised by the same multiplicative seasonality factor, we get:

$$\frac{S_t}{S_{t-1}} = \frac{sS_{a,t}}{sS_{a,t-1}} = \frac{S_{a,t}}{S_{a,t-1}}$$

where s represents the multiplicative seasonality coefficient, common to both prices, $S_{a,t}$ corresponds to the deseasonalised price for period t . When we form the price ratio based on different contracts, there is no guarantee that the seasonality coefficients will cancel. Finally, we need to keep in mind that many energy commodities trade for 6 days during a week (for example, an on-peak period in California is defined as Monday through Saturday).

3.2.2 Estimation of Volatility for a Mean Reverting Process

For illustration in this section we assume a simple Ornstein-Uhlenbeck process for a particular price (spot or forward) S of the form

$$dS = \alpha(\bar{S} - S)dt + \sigma dz \quad (3.2)$$

If the starting price level is $S(0)$, the expected price at time t will be given by $\bar{S} + (S(0) - \bar{S})\exp(-\alpha t)$. As the term $\exp(-\alpha t)$ goes to 0, as t becomes large, the long-term expectation of price is equal to $\bar{S}$.

One should observe that the second term in the equation (3.2) is similar to the diffusion part of GBM, but that the interpretation of volatility changes. The change in price, dS , is measured in dollars per physical unit, dz , is unitless, and therefore σ in equation (3.2) must be measured in dollars as well, unlike the volatility used in the Black-Scholes-Merton option model. The important lesson here is that we define and estimate volatility in the context of the stochastic process assumption and when this changes, the interpretation of volatility changes as well.

Volatility can be equally easily estimated from the historical data, if a mean reversion process is assumed. A spreadsheet-friendly technology that jointly estimates the mean reversion coefficient (c.f. section 2.9) and the volatility parameter, is based on discretisation of the continuous stochastic process given by equation (3.2) into the following equation that represents an autoregressive process of order one:

$$\Delta S_t = \alpha_0 + \alpha_1 S_t + \varepsilon_t \quad (3.3)$$

This follows from the observation that equation (3.2) is the limiting case of the following process (as $dt \rightarrow 0$):

$$S_{t+1} - S_t = \bar{S}(1 - e^{-\alpha}) + (e^{-\alpha} - 1)S_t + \nu_t \quad (3.4)$$

The error term ν_t in (3.4) is normally distributed with variance given by:

$$\sigma_\nu^2 = \frac{\sigma^2}{2\alpha}(1 - e^{-2\alpha}) \quad (3.5)$$

The parameters of the original equation (3.2) can be recovered from the estimated coefficients of the discrete version:

$$\bar{S} = -\alpha_0/\alpha_1 \quad (3.6)$$

$$\alpha = -\ln(1 + \alpha_1) \quad (3.7)$$

$$\sigma = \sigma_\nu^2 \sqrt{\frac{\ln(1 + \alpha_1)}{\ln(1 + \alpha_1)^2 - 1}} \quad (3.8)$$

3.2.3 Volatility Estimation: Special Issues

This section contains the discussion of several special issues: the choice of the annualisation factor and the use of intra-period data (typically this means intraday prices).

A typical textbook discussion of the annualisation factor typically revolves around the correct number of trading days in a year. However, the issue is more complicated than this. French and Roll (1986) compared the variance of price returns during the week with the variance of price returns calculated for the weekend. One can calculate four daily price returns for weekdays (Monday over Tuesday, Wednesday over Tuesday, etc) and one weekend return (Monday close to Friday close). For the US stock data used in their study the authors determined that the weekend was equivalent to 1.107 trading days. This result can be easily explained. Volatility is related to trading activities that are in turn triggered by the flow of information into the trading rooms. Over the weekend, the information flow slows down when trading is suspended and new information is processed during trading activity on Monday. This explains why the three calendar day period is practically equal to one trading day. Using this finding one can argue for an annualisation factor of 266 [= 52 × (4 + 1.107)].

However, energy markets are different. In many cases the information flow does not slow down significantly on weekends. One can argue that the information regarding political events and weather (which is extremely important for the natural gas, heating oil, propane and electricity markets) keeps flowing roughly at the same rate over the weekend as during the working week. Data analysis confirms our intuition. We have looked at the weekday/weekend volatility for natural gas and crude oil using the prompt month NYMEX contracts yearly from 1995 until the first quarter of 2000. The results are summarised in Table 3.1.

TABLE 3.1 Weekday/Weekend Volatility Ratio (%)

Year	Natural Gas	Crude
1995	103.41	119.81
1996	134.57	138.05
1997	128.72	113.98
1998	125.03	144.51
1999	115.47	88.17
2000	152.78	93.51
Average	125.02%	125.90%

As we can see, the ratio of weekend to weekday volatility may be significantly higher than in the case of equities. On average, the volatility ratio implies the annualisation factor of the square root of 273 [= $52 \times (4 + 1.25)$].

What can be done in view of these findings? One natural solution is to agree on the annualisation factor that will be used across the board for all commodities. This guarantees that any volatility quote will be unambiguous and can be restated by another trader based on his personal views of the market. Our recommendation is to use $\sqrt{250}$.

3.2.4 Intraday Price Variability

Volatility is typically estimated from the market closing prices. In many cases, using closing price information for volatility estimation causes the loss of very valuable information as additional information is available about the price changes between the market opening and closing prices. Many markets, and energy markets in particular are characterized by very high volatility of intraday prices.

3.2.5 Estimation of Volatility For a Basket

In many energy markets, both producers and end-users hedge their price exposures with so-called basket options. A basket option is defined in terms of a sum of two or more prices. In some cases, the prices may be weighted using weights corresponding to the option buyer's economic sensitivity to different component prices. For example, an oil and natural gas producer may be interested in buying an option on a composite commodity that is constructed by weighting oil and gas prices (expressed in terms of a common BTU equivalent) by 0.7 and 0.3, respectively. To be more precise, the call option payoff is defined as

$$\max(0, w_1 S_1 + w_2 S_2 - K) \quad (3.9)$$

and the put option payoff is defined as

$$\max(0, K - (w_1 S_1 + w_2 S_2)) \quad (3.10)$$

where S_1 and S_2 stand for the prices of the oil and gas and the w 's stand for the weights. These options can be seen as options written on a composite energy with the volatility derived from the volatilities of the contributing commodities. The question is how to derive this composite volatility in a formal way. The answer is produced through the application of Ito's lemma.

Assume that the prices of the two commodities follow GBM processes given by

$$dS_1 = \alpha_1 S_1 dt + \sigma_1 S_1 dz$$

and

$$dS_2 = \alpha_2 S_2 dt + \sigma_2 S_2 dw$$

where dz and dw stand for correlated (with a correlation coefficient of ρ) Brownian motions. The multivariate extension of Ito's lemma to some function $G = G(S_1, S_2, t)$ is given by:

$$dG = \frac{\partial G}{\partial S_1} dS_1 + \frac{\partial G}{\partial S_2} dS_2 + \frac{\partial G}{\partial t} dt + \frac{1}{2} \left(\frac{\partial^2 G}{\partial S_1^2} dS_1^2 + \frac{\partial^2 G}{\partial S_1 \partial S_2} dS_1 dS_2 + \frac{\partial^2 G}{\partial S_2^2} dS_2^2 \right) \quad (3.11)$$

Applying Ito's lemma to $S_1 + S_2$ renders

$$\sigma_{S_1+S_2} = \frac{\sqrt{\ln[S_1^2 e^{\sigma_1^2 t} + 2S_1 S_2 e^{\rho \sigma_1 \sigma_2 t} + S_2^2 e^{\sigma_2^2 t}] - 2 \ln[S_1 + S_2]}}{\sqrt{t}} \quad (3.12)$$

3.2.6 Implied Volatility

The previous sections have dealt with estimation of volatility from historical data. The job of an option trader is to guess what is the level of volatility expected in the future. A buyer or a writer of an option that is unhedged effectively makes a bet on the price level of the underlying instrument. An option trader who hedges makes a bet on volatility. Volatility that is used as an input to an option pricing formula that equates the model price with the market price is called an implied volatility. It is implied in the sense that once an option market price is known one can use the option pricing formula to calculate the volatility input, given that other inputs are known (maturity, the current price of the underlying, interest rate, the strike price). Volatility can be 'backed out' from market prices using a trial and error method or a program allowing for a solution of a nonlinear equation (an option pricing equation in this case). In practice, implied volatility is usually quoted for at-the-money options and is often calculated based on the average of an at-the-money straddle (a call and a put with the same strike price).

It is well-known fact that out-of-the-money options trade at volatility levels that diverge from at-the-money volatility. This is often referred to as a volatility smile (or a frown, smirk, etc. depending on the shape of the curve). The traders try to compensate for the shortcomings of the model by adjusting the inputs to the model. Given that most inputs are well defined and directly observable and cannot be adjusted at will, this leaves

only one free parameter that can be manipulated. The adjustment would lead typically to a volatility smile – the traders increase the volatility for out-of-the-money options to compensate for flat volatility model derived valuations that are perceived to be too low. There are many competing explanations as to why this is the case, which are not necessarily mutually exclusive.

The first explanation is related to well-known shortcomings of the Black-Scholes-Merton model which is based on the assumption of normality of price returns and as such understates the probability of extreme outcomes. One consequence of jumps, in for example electricity prices, which is important from the point of view of management of portfolios of power derivatives, is the presence of so-called *fat tails*. This term is used to refer to the fact that the probability of extreme outcomes inferred from the empirical data exceeds theoretical probabilities which are based on probability distributions typically assumed in the model. Distributions displaying fat tails are referred to as *leptokurtic*. This property is measured by the kurtosis, or the fourth moment around the mean, which for the normal distribution is equal to three (in some computer programs, this number is adjusted to be equal to 0). Table 3.2 displays calculated values of excess kurtosis for a number of price indices in the US power markets to illustrate the prevalence of leptokurtic distributions in the electricity markets. The program used for this calculation produces zero kurtosis for the normal distribution. Hence, we use the traditional term “excess kurtosis”.

One should notice that there are several competing explanations of the presence of fat tails in empirical price distributions. One source of fat tails is the presence of jumps in the prices which, as we have seen, are a typical feature of power markets. Another explanation is that the parameters of the distribution of returns (which is usually assumed to be normal) vary over time. Empirical observations of returns are drawn in this case from a mixture of distributions.

This feature of power markets creates not only a challenge from the point of view of pricing electricity derivatives but also from the point of view of portfolio management and *value-at-risk* calculations. Value-at-risk is a shorthand term for a system that measures the potential loss in the value of a portfolio of financial instruments over a specified time period, typically one day, with a given probability, and a subject which is covered in some depth in chapter 10. For portfolios that include electricity derivatives, incorporation of gapping prices is of critical importance to insure a necessary degree of realism¹⁴. The degree to which empirical distributions may diverge from theoretical distributions can be assessed by comparing the third and fourth moments of the empirical distributions with the theoretical levels (both equal to zero for a normal distribution).

TABLE 3.2 Excess Kurtosis of Selected Power Prices, 29/6/1995–2/10/1997

Location	PJM	Palo Verde	COB
Kurtosis	4.87	14.48	10.78

An alternative explanation for volatility smiles deals with the microstructure of the markets. In many emerging energy markets, insufficient liquidity or lack of complete markets does not allow for effective delta hedging. This means that the supply of options comes either from the trading institutions with exceptionally efficient trading operations or from the organizations that can mitigate the risk of financial options using positions in physical assets. For example, power generators can sell call options on electricity, treating them as covered calls: if an option is exercised an idle unit may be turned on to deliver into the contract. The buyers see out-of-the money options as disaster insurance. This reduces the supply of options relative to demand and increases their prices. This is reflected in turn in implied volatility higher than the volatility for at the money options.

3.3 Stochastic Volatility Models

In section 2.5 we introduced the concept of stochastic volatility models for energy price evolution. Recall that a simple model that yields skewed spot price return distributions has a stochastic term that can be generalised as $\sigma S^\gamma dz$, where γ is between 0 and 1. Any dependence on price is factored out, and the volatility term σ is explicitly constant with respect to time. A process with constant volatility is termed homoskedastic, while a process with time dependent volatility, be it deterministic or random, is termed heteroskedastic. Part of the appeal of constant-volatility models is their mathematical tractability and relative ease of estimation. There is no mathematical reason, however, that requires volatility to be a constant, and indeed many commodity markets exhibit time varying volatility. For example, figure 3.1 shows the 30-day rolling volatility for Californian average daily PX prices from April 1, 1998 until May 31, 2000, and clearly suggests a price volatility that is not constant in time but rather is best described by a process with both periodic and stochastic elements.

We have already seen a simple example of stochastic volatility in section 2.5 with the Hull-White mean reverting model represented by equations (2.10) and (2.11), where the variance is itself cast as a diffusion process. Price models with deterministically time dependent volatility are treated in chapter 8 in the context of forward contract price dynamics.

As noted in the previous section, energy commodity price return distributions tend to be leptokurtic or fat tailed. In addition, some commodities, such as crude oil, also exhibit volatility clustering. These behaviours are consistent with a process where the volatility is both stochastic and autoregressive. Models that attempt to incorporate and explain these behaviours go by the generic name of Autoregressive Conditional Heteroskedastic or, less formidably, simply ARCH models. First proposed by Engle (1982) to model price inflation, ARCH models have been successfully applied in many different areas of finance including interest rates, foreign exchange, and equities. As a result, variations on the basic concept are myriad and the literature on the subject is extensive. See Bollerslev (1992) for a review.

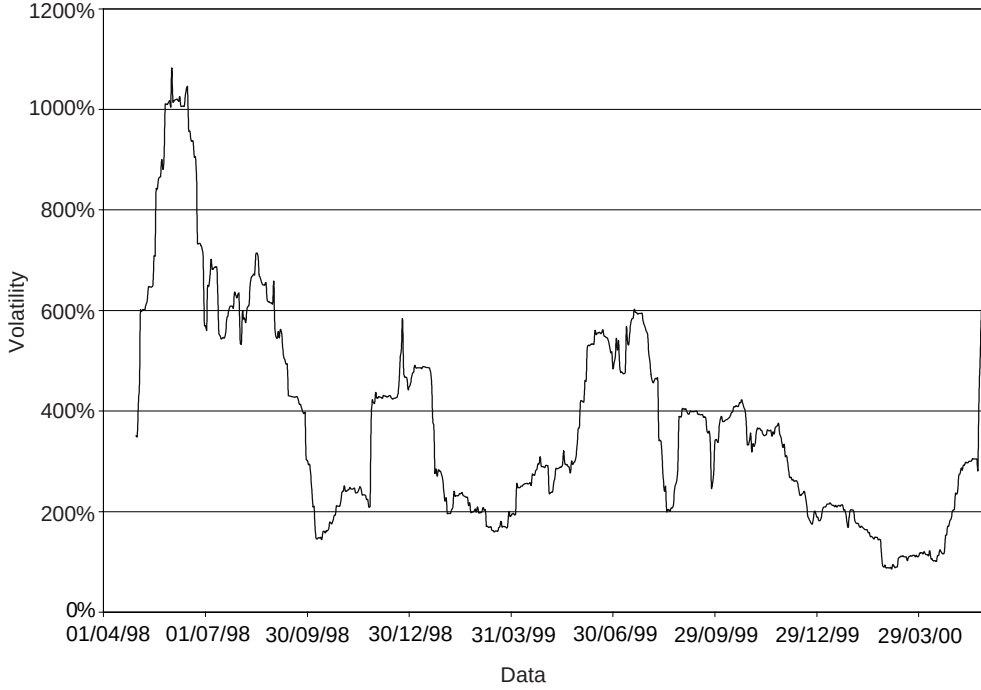


FIGURE 3.1 30-Day Rolling Annualized Volatility. Californian Day-Ahead Average PX Price (April 1, 1998 - May 31, 2000)

It is typical in ARCH models of asset prices to write the price returns as,

$$y_t = \ln\left(\frac{S_t}{S_{t-1}}\right) = \kappa + u_t \quad (3.13)$$

where κ is a constant mean term, and u_t is the “innovation at time t ”. The innovations are stochastic and written as $u_t = \sigma_t \varepsilon_t$, where ε_t is an independent random error distributed as $N(0,1)$. Now the essential feature of ARCH modelling is to make the innovation variance at time t conditional on previously realised innovations. For example in an ARCH(q) model, the variance would be represented as;

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \alpha_2 u_{t-2}^2 + \dots + \alpha_q u_{t-q}^2 \quad (3.14)$$

Of course, the order of the equation q is arbitrary. In practical applications however, models with several terms are often required in order to capture the ‘decaying memory’ of volatility. To incorporate this feature in a more parsimonious way, Bollerslev (1986) developed a generalised ARCH model, or GARCH(p,q) model. Here the variance equation includes both past innovations (of order q) and past conditional variances (of

order p):

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 \quad (3.15)$$

Note that because the equation is recursive in variance, σ_t will effectively depend on all innovations. As a result, a GARCH model with a few terms will often fit a data set as well as an ARCH model of much higher order. In practice, the first order GARCH(1,1) model, shown in equation (3.16), is often quite adequate.

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad (3.16)$$

Figures 3.2a and b show an example of a spot price process with a GARCH(1,1) volatility structure. Figure 3.2a shows the price path and associated rolling thirty day volatility, and Figure 3.2b shows the frequency distribution of the log price returns together with a superimposed normal distribution, the mean and standard deviation of which have been derived from the data. As expected, prices exhibit periods of high and low volatility, and the distribution of log returns is leptokurtic.

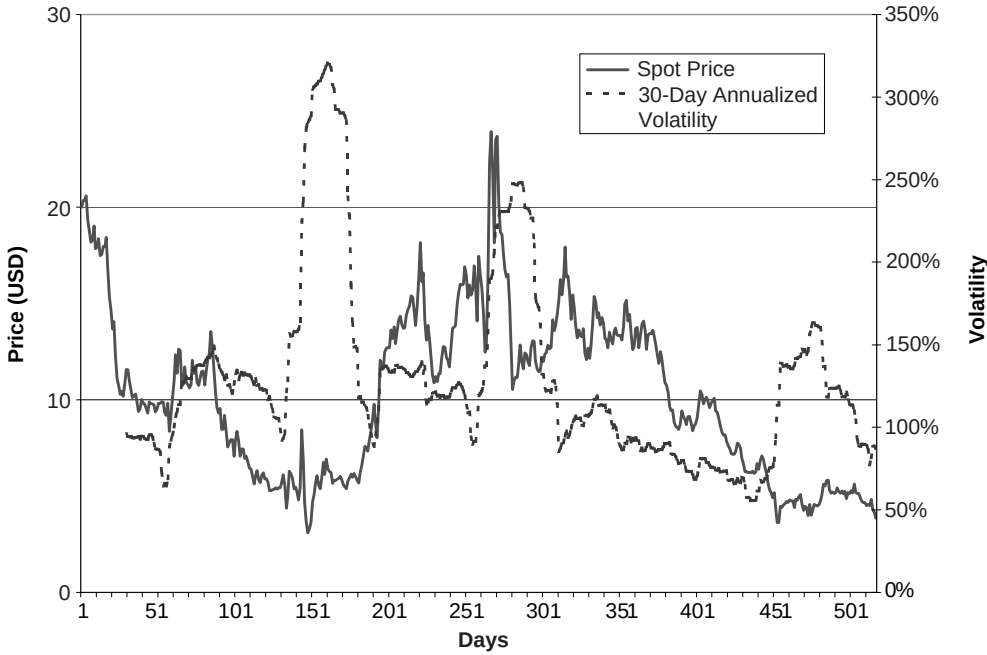


FIGURE 3.2A Price Process and 30-day Rolling Annualised Volatility modelled using eq. 3.13 with GARCH(1,1) volatility (as described in eq. 3.16). Model parameters: $\kappa = 0.0$, $\alpha_0 = 0.0006$, $\alpha_1 = 0.30$, $\beta_1 = 0.60$

3.3.1 Estimation and Testing

Once a model is chosen as a hypothesis for a given price behaviour, the parameters for that model must be estimated. In addition to parameter estimation, an equally important and often times more difficult task is to test how well the proposed model describes the data. The fields of estimation and testing are vast, representing core subjects in statistics and econometrics. By necessity we will be brief in this section, concentrating only on the essential aspects of estimation as it relates to energy commodity price modelling. More extensive discussions of the topics touched on here may be found in Greene (1997) and Davidson and MacKinnon (1993), for example.

As we shall see in a moment, there is more to parameter estimation than simply running regressions. There are an infinite number of possible estimators for a given model parameter: an arbitrary number, 42 for example, is a perfectly acceptable estimator of the mean of a price process; it is simply not very accurate, and hence not very useful. Two properties, bias and efficiency define the usefulness of a parameter estimator. An estimator of a parameter is a random number with a mean and variance. Clearly, we would like our measurement to be ‘the best guess’ possible of the actual parameter value. In other words, if $\tilde{\theta}$ is the estimate of the true value θ , then it is desirable that $E[\tilde{\theta} - \theta] = 0$. An estimator is said to be unbiased if this property holds. Similarly, it is preferable to have an estimator with the smallest possible variance. For example, let θ_1

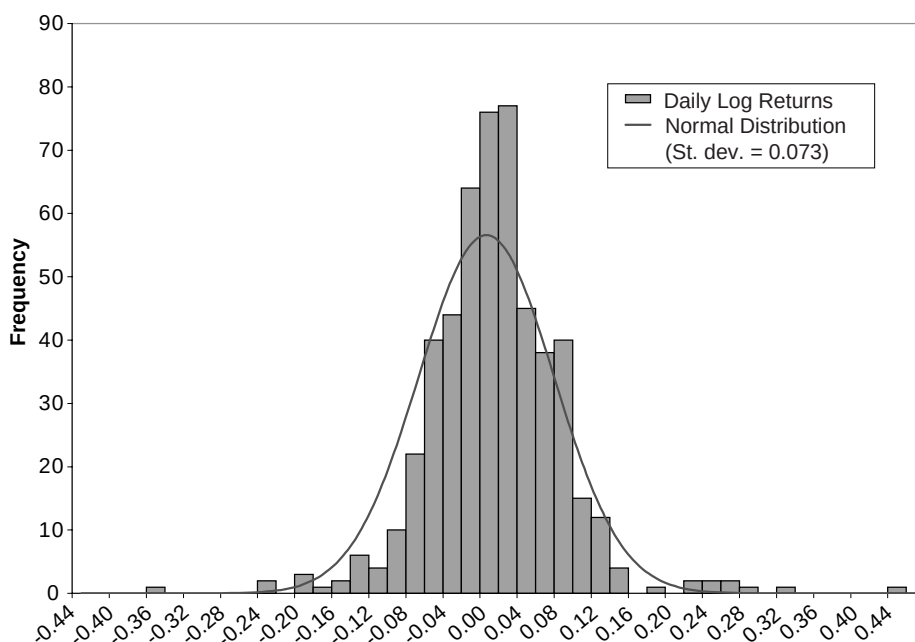


FIGURE 3.2B Histogram of the Log of Daily Price Returns

be the average of the sum of every fourth price in a time series. This is an entirely acceptable and unbiased estimate of the mean price. However, the variance of θ_1 is twice as large as that of the estimate θ_2 derived from taking the average of the sum of every price. Clearly, θ_2 is more 'efficient' than θ_1 because, $E(\theta_1^2) > E(\theta_2^2)$.

Both properties are important. An estimator that is biased but efficient may be preferable to an unbiased, inefficient estimator because it will provide tighter bounds on the range of possible parameter values.

For the sake of completeness, we mention in passing that the concepts of bias and efficiency are applicable to finite-sample estimates only. It is often the case that we do not know the finite properties of an estimator. Therefore we must instead compare the asymptotic (i.e. in the limit of large sample size) properties of estimators to determine which is more suitable and then assume that the suitability holds in the real world of finite sampling. The asymptotic properties most often described in the literature as desirable are consistency, asymptotic normality, and asymptotic efficiency. While consistency and asymptotic normality are akin to bias in that they measure if an estimate converges in some sense to the true value, they are not the same things. Indeed, an estimator can be biased but consistent, or vice versa.

3.3.2 Ordinary Least Squares

Ordinary Least Squared estimation (OLS) is the technique most likely to be familiar to readers because it is at the heart of standard linear regression analysis. The concept is simple: we hypothesise that a set of N observations $\{y_t, \mathbf{x}_t\}$ is related via the stochastic equation

$$y = \boldsymbol{\theta} \cdot \mathbf{x} + \varepsilon \quad (3.17)$$

where $\boldsymbol{\theta}$ is the set of unknown parameters. We can estimate these parameters by choosing a set $\tilde{\boldsymbol{\theta}}$ such that the sum of squared observed residuals (SSR) is minimised;

$$y_t = \tilde{\boldsymbol{\theta}} \cdot \mathbf{x}_t + e_t$$

$$SSR = \sum_{t=1}^N e_t^2 = \sum_{t=1}^N (y_t - \tilde{\boldsymbol{\theta}} \cdot \mathbf{x}_t)^2 \quad (3.18)$$

The OLS estimates $\tilde{\boldsymbol{\theta}}$ are found in the usual way by setting $\frac{\partial SSR}{\partial \boldsymbol{\theta}} = 0$ and solving for $\tilde{\boldsymbol{\theta}}$ ¹. For example, suppose that we have a set of N observations $\{y_t\}$ that we hypothesise is the result of the pure diffusion equation represented above in equation (3.13):

$$y_t = \kappa + \sigma \varepsilon_t, \quad \varepsilon_t \sim N(0, 1) \quad (3.19)$$

¹ Of course, one must also examine the second derivative to ensure that the solution represents a maximum.

The sum of squared residuals of this set is then

$$SSR = \sum_{t=1}^N e_t^2 = \sum_{t=1}^N (y_t - k)^2 \quad (3.20)$$

Taking the derivative with respect to k and setting the resultant equation equal to zero yields;

$$\begin{aligned} \frac{\partial SSR}{\partial k} &= \sum_{t=1}^N 2(y_t - k) = 0 \\ k &= \frac{1}{N} \sum_{t=1}^N y_t \\ \kappa &= E(k) = \frac{1}{N} \sum_{t=1}^N y_t \end{aligned} \quad (3.21)$$

as expected. Because σ multiplies the stochastic term, we must resort to a slightly different method to find its estimate. From equations (3.18) and (3.19) we have that

$$\begin{aligned} \sigma \varepsilon_t &= y_t - \kappa \\ e_t &= y_t - k \end{aligned} \quad (3.22)$$

We use these relations to define an *average* innovation and *average* return;

$$\bar{y} = \frac{1}{N} \sum_{t=1}^N y_t = k \quad (3.23)$$

and

$$\sigma \bar{\varepsilon} = \sigma \frac{1}{N} \sum_{t=1}^N \varepsilon_t = \bar{y} - \kappa \quad (3.24)$$

This in turn allows us to make the following substitutions;

$$\begin{aligned} e_t &= y_t - k \\ &= \kappa + \sigma \varepsilon_t - k \\ &= \sigma(\varepsilon_t - \bar{\varepsilon}) \end{aligned} \quad (3.25)$$

Squaring equation (3.25) gives

$$\begin{aligned} e_t^2 &= \sigma^2(\varepsilon_t - \bar{\varepsilon})^2 \\ &= \sigma^2(\varepsilon_t^2 - 2\varepsilon_t\bar{\varepsilon} + \bar{\varepsilon}^2) \end{aligned} \quad (3.26)$$

which, when summed over all N observations, yields;

$$\sum_{t=1}^N e_t^2 = \sigma^2 \left(\sum_{t=1}^N \varepsilon_t^2 - N\bar{\varepsilon}^2 \right) \quad (3.27)$$

Now the random variables ε_t are distributed normally as $N(0,1)$ and are independent, which implies that $E(\varepsilon_i \varepsilon_j) = \delta_{ij}$. Therefore, we arrive finally at

$$\sum_{t=1}^N e_t^2 = \sigma^2(N-1)$$

$$\sigma = \sqrt{\frac{1}{N-1} \left(\sum_{t=1}^N e_t^2 \right)} \quad (3.28)$$

which, of course, is the common definition for standard deviation. If the stochastic equation (3.19) had p parameters rather than just one, it can be shown through a similar analysis that the OLS estimate for σ in this case would be equation (3.28) with the denominator $N-1$ replaced by $N-p$ as expected.

OLS estimators have the desirable properties of unbiasedness, consistency, and efficiency for a variety of problems. However, in the case of heteroskedastic systems, OLS estimators are not generally efficient, therefore other methods are often employed to provide estimates.

3.3.3 Maximum Likelihood

We introduced maximum likelihood in section 2.11 as a powerful and general estimation method. In this section, we elaborate on the principle underlying the method by again using the example of the pure diffusion process represented by equation (3.13).

$$y_t = \kappa + \sigma \varepsilon_t$$

Recall that ε_t is distributed as $N(0,1)$, which implies that y_t is distributed as $N(\kappa, \sigma)$. The probability density function for each y_t is then,

$$p(y_t; \kappa, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y_t - \kappa}{\sigma}\right)^2} \quad (3.29)$$

Now, every y_t corresponds to an independent observation, so we can write the joint distribution for a set of, say N , observations as,

$$p(\{y_t\}; \kappa, \sigma) = \prod_{t=1}^N \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y_t - \kappa}{\sigma}\right)^2} \quad (3.30)$$

We can think of this function as representing the probability or likelihood that we would draw exactly this set of $\{y_t\}$. For this reason, the joint density is typically called the likelihood function and is written as

$$L(\Theta; \{y_t\}) = p(\{y_t\}; \Theta) \quad (3.31)$$

emphasising that its value, given a set $\{y_t\}$ of observations, is a function of the

distribution's parameter set Θ . With this definition, we can now turn the process around. We now hypothesise that this set of numbers is drawn from a normal distribution of unknown mean and variance and ask the following question. From what distribution is it most likely that these numbers were drawn, or, put another way, which model parameters (the values of κ and σ^2 in the case of equation (3.30)) define the distribution that has the maximum likelihood of producing this set of numbers? The answer is found by maximising the likelihood function with respect to the model parameters. This is accomplished in the usual way by setting the gradient with respect to the regression parameters equal to zero and solving for each parameter value. For equation (3.30) the maximising equations are

$$\ln L(\kappa, \sigma; \{y_t\}) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln \sigma^2 - \frac{1}{2} \sum_{t=1}^N \left[\frac{(y_t - \kappa)^2}{\sigma^2} \right] \quad (3.32a)$$

$$\frac{\partial \ln(L(\kappa, \sigma; \{y_t\}))}{\partial \kappa} = \frac{1}{\sigma^2} \sum_{t=1}^N (y_t - \kappa) = 0 \quad (3.32b)$$

$$\frac{\partial \ln(L(\kappa, \sigma; \{y_t\}))}{\partial \sigma} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{t=1}^N (y_t - \kappa)^2 = 0 \quad (3.32c)$$

Note the use of the log likelihood function $\ln L(\kappa, \sigma; \{y_t\})$. This is a common practice because the natural log is typically easier to treat mathematically and computationally, and parameters that maximise the log likelihood are guaranteed to maximise the likelihood function as well.

The solutions to equation (3.32) are referred to as the Maximum Likelihood Estimates or MLEs of the true distribution parameters. An obvious question is whether or not MLEs are the “best guesses” of the true parameter values. We state without proof that, assuming the likelihood function obeys some specific regularity conditions, ML estimators are consistent, asymptotically normal, and asymptotically efficient. In fact, ML estimators are at least as asymptotically efficient as any consistent estimator.

It is interesting to note that although ML estimation is consistent, it is not unbiased in general. For example, the MLE for the variance derived from equation (3.32c) is found to be;

$$\begin{aligned} \frac{\partial \ln(L(\kappa, \sigma; \{y_t\}))}{\partial \sigma} &= -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{t=1}^N (y_t - \kappa)^2 = 0 \\ \frac{1}{2\sigma^4} \sum_{t=1}^N (y_t - \kappa)^2 &= \frac{N}{2\sigma^2} \\ \sigma^2 &= \frac{1}{N} \sum_{t=1}^N (y_t - \kappa)^2 \end{aligned} \quad (3.33)$$

which is not the same as the OLS unbiased estimate

$$\sigma^2 = \frac{1}{N-1} \sum_{t=1}^N (y_t - \kappa)^2.$$

As we mentioned before however, the ML estimate is more efficient than OLS and, in cases of small numbers of observations, may be a better estimate in the sense of having the minimum mean squared error.

3.3.4 Testing

Having decided on a hypothesis for the price process under study and estimated the model's parameters, it is essential to test the validity of that hypothesis. Unfortunately, the importance of this step is sometimes ignored in the rush to produce results. For example, the ubiquity of Black-Scholes-Merton option pricing in equity markets means that it is very common to assume that pure diffusion also describes energy commodity prices. Figure 3.3 shows average daily prices for NSW electricity for the calendar year 1999.

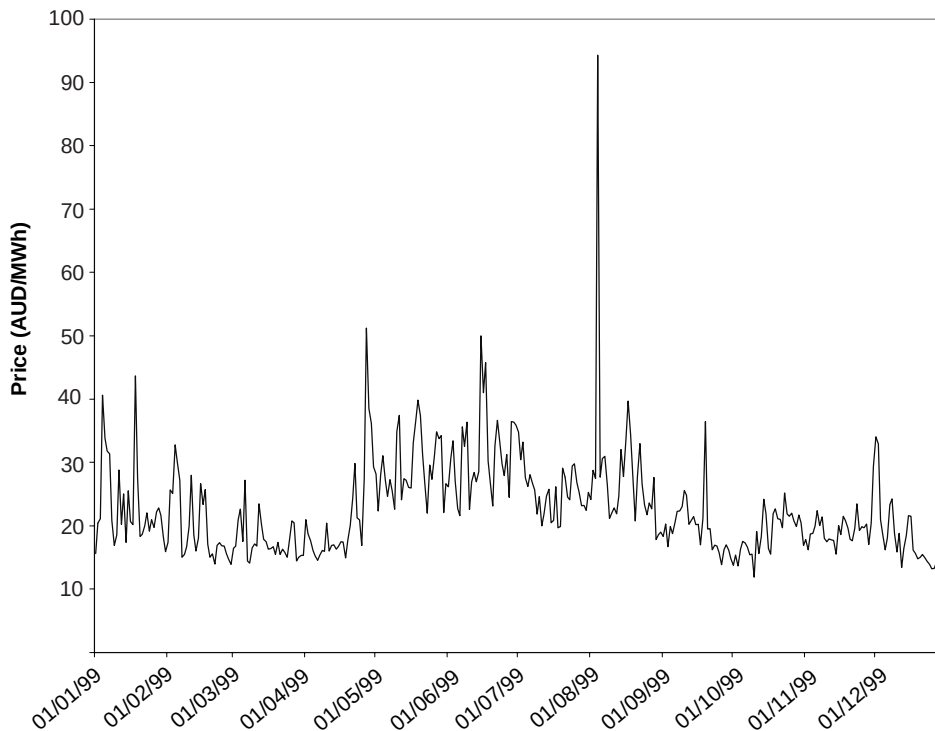


FIGURE 3.3 Daily Average NSW Power Prices (Jan. 1, 1999–Dec. 31, 1999)

The pure diffusion variance estimate, described earlier in this chapter, yields a daily volatility of 21.4%. Without other test statistics, however, it is impossible to judge how meaningful this number is and how well the diffusion process explains the price behaviour. We can make qualitative judgements by graphing the frequency of daily returns and overlaying a Gaussian curve with zero mean and standard deviation equal to the derived volatility as shown in figure 3.4.

This figure demonstrates the poor quality of the fit. The standard deviation seems to be overly influenced by a few large valued returns at the expense of a larger number of small valued returns. Indeed, fitting “by eye”, as suggested by the dashed normal distribution in the figure, might lead us to estimate a rather lower volatility. Calculating the skew and excess kurtosis of the return distribution provides a more quantitative test of normality. If the returns are normally distributed, then the two statistics

$$T_{\text{skew}} = \frac{1}{\sqrt{6N}} \sum_{i=1}^N \left(\frac{y_i - \kappa}{\sigma} \right)^3 \quad (3.34)$$

and

$$T_{\text{kurt}} = \frac{1}{\sqrt{24N}} \sum_{i=1}^N \left[\left(\frac{y_i - \kappa}{\sigma} \right)^4 - 3 \right] \quad (3.35)$$

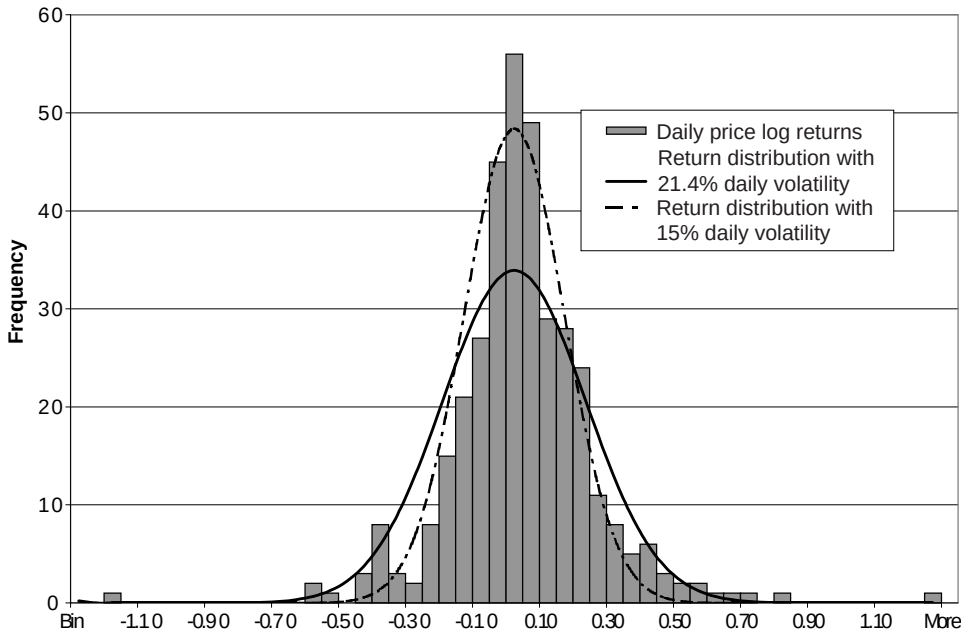


FIGURE 3.4 Daily Average NSW Power Prices Log Returns (Jan. 1, 1999–Dec. 31, 1999)

are asymptotically distributed as $N(0,1)$. A related statistic, suggested by Jarque and Bera (1980), essentially combines equations (3.34) and (3.35) to form

$$T_{JB} = N \left[\frac{\text{Skew}^2}{6} + \frac{\text{Excess Kurtosis}^2}{24} \right] \quad (3.36)$$

The Jarque-Bera statistic has a chi-squared distribution of degree 2 under the null hypothesis of normality. In the case of the NSW daily price data, we find for the 364 observations the skew and excess kurtosis to be 0.177 and 6.22 respectively. The statistic T_{JB} is then about 590 implying that we reject the hypothesis of normally distributed log price returns at a level of order 10^{-128} .

Tests for the presence of heteroskedasticity rely on the fact that the residuals from OLS regression will retain any non-stationary behaviour inherent in the underlying process. Thus, tests sensitive to changing variance can be performed on the regression residuals to detect heteroskedasticity. One such time-honoured test is the Goldfeld—Quandt test. It assumes that there exists a variable that is related to a model's variance. Using this variable, the data is sorted into subsets of high and low expected variance. We then perform OLS regression on the two data sets and form the statistic,

$$T_{GQ} = \frac{SSR_1/(n_1 - K)}{SSR_2/(n_2 - K)} \quad (3.37)$$

Where SSR stands for Sum of Squared Residuals, n_1 and n_2 are the number of observations in the high and low data sets respectively and K is the number of parameters in the regression. Under the null hypothesis of homoskedasticity, this statistic is distributed as $F[n_1-K, n_2-K]$.

To test specifically for ARCH and GARCH behaviour, Engle (1982) and Bollerslev (1986) suggest variations on the Breusch—Pagan/Godfrey test, which is a form of Lagrange multiplier test. The details of this test are beyond the scope of this chapter, but the interested reader may refer to standard econometric texts such as Davidson and MacKinnon (1993) for a discussion of the Lagrange multiplier principle.

A related test is suggested by Hull (1999), which can be used to measure the success of a GARCH model fit. If the model explains the data well, then the set of $\{u_t^2/\sigma_t^2\}$ should exhibit no autocorrelation. Here u_t and σ_t are the residuals and volatilities derived from the GARCH model under examination. The Ljung-Box statistic is suggested as a method for testing this series for autocorrelation.

3.3.5 Examples from the Energy Commodity Markets

We conclude this chapter by comparing different price models fitted to selected energy markets: the California PX day-ahead peak electricity price for zone NP15 (NP15), the NYMEX prompt month gas futures contract (Gas), and the NYMEX prompt month crude oil futures contract (Crude). Table 3.3 details basic statistics for each data set used in this analysis.

TABLE 3.3 Data set statistics for daily log price returns of California PX NP-15 peak prices, NYMEX prompt month gas futures contract prices, and NYMEX prompt month crude oil futures contract prices

	<i>NP15</i>	<i>Gas</i>	<i>Crude</i>
Span	January 15 th , 1999 – March 29 th , 2000		
Number of Observations*	370	303	303
Mean	0.68×10^{-3}	1.61×10^{-3}	2.57×10^{-3}
Variance	0.23	8.52×10^{-4}	5.29×10^{-4}
Skew	0.46	-0.07	-0.51
Excess Kurtosis	11.98	2.00	1.10

* California peak prices are quoted six days per week excluding holidays, while NYMEX commodity contracts are traded only five days per week excluding holidays.

TABLE 3.4 Model Specifications

	<i>Return process</i>	<i>Volatility process</i>	<i>Equation Reference</i>
Pure Diffusion (PD)	$y_t = \kappa + \sigma \varepsilon_t$	σ constant	3.13
Jump Diffusion (JD)*	$y_t = \kappa + \sigma \varepsilon_t$	σ constant	
Diffusion + ARCH(1) (D + Arch)	$y_t = \kappa + \sigma_t \varepsilon_t$	$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2$	3.14
Diffusion + GARCH(1,1) (D + Garch)	$y_t = \kappa + \sigma_t \varepsilon_t$	$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2$	3.16
Jump Diffusion + GARCH(1,1) (JD + Garch)*	$y_t = \kappa + \sigma_t \varepsilon_t$	$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2$	

* In the case of Jump Diffusion, the last term is the product of J , a Bernoulli distributed random variable with a probability ϕ of being one and probability $(1-\phi)$ of being zero, and q , a random number distributed as $N(\mu, \gamma^2)$.²

We estimated the parameters for five different models, outlined in Table 3.4, following maximum likelihood methods similar to those described by Jorion (1988) and Das (1995).

In general, we employed the “BHHH” algorithm developed by Berndt, Hall, Hall, and Hausman (1974) to perform the regressions. The BHHH technique is essentially a

² The use of a Bernoulli random number follows the work by Ball and Torous (1983) and more recently Das (1995). It is well known that the Bernoulli process is a stable approximation of the Poisson process and converges to it in the limit. This methodology differs from that employed by Jorion (1988), which involves truncation of the infinite summation of the Poisson process. In essence, the likelihood function is cast as a mixture of two normals (for a detailed discussion, see Akgiray and Booth (1987)); namely,

$$L(\kappa, \sigma, \mu, \gamma; y_t) = P(J=0)P(y_t|J=0) + P(J=1)P(y_t|J=1) \\ = (1-\phi) \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{(y_t - \kappa)^2}{2\sigma_t^2}\right) + \phi \frac{1}{\sqrt{2\pi(\sigma_t^2 + \gamma^2)}} \exp\left(-\frac{(y_t - \kappa - \mu)^2}{2(\sigma_t^2 + \gamma^2)}\right)$$

gradient search method appropriate for maximising the log likelihood. Although usually robust, we found the BHHH method failed to converge for some data sets and model choices. In some cases, we were able to achieve convergence by first using a standard Simplex search of the parameter space to determine the location of the global maximum. The results of the Simplex search were then used as starting points for the BHHH algorithm.

The relative goodness of fit for different models can be compared using the likelihood ratio test. Under the null hypothesis that the enlarged parameter space does not improve the fit to the data, the ratio test statistic is defined as $-2\Delta L(\Theta; y)$, where $\Delta L(\Theta; y)$ is the difference between the two log likelihood function values. This test statistic has a chi-square distribution with degrees of freedom equal to the difference in the number of parameters estimated between the two spaces. Another related measurement of fit is the Schwarz criterion (Schwarz (1978)) defined as:

$$SC = -2L(\Theta; y) + k \ln N \quad (3.38)$$

where $L(\Theta; y)$ represents the likelihood function, k is the number of parameters to be estimated and N is the number of observations. The Schwarz criterion accommodates the trade-off between better fit and more parameters by penalising the model with larger number of parameters. The most probable model is the one with the smallest Schwarz criterion. Table 3.5 shows the Maximum Likelihood results of the fitting procedure together with Schwarz criteria for each model and commodity.

A number of interesting conclusions can be inferred from the data in Tables 3.3 and 3.5. The electricity data are highly skewed and leptokurtic. The Jarque-Bera statistic described in equation (3.36) is about 2000 for the electricity data, which overwhelmingly rejects the hypothesis of normality. Also interesting are the results for gas and crude oil. Over this time period, both return series exhibited platykurtic, or thin-tailed behaviour, and therefore, as might be expected, the ARCH type models did not outperform the pure diffusion fit. The electricity data, in contrast, appears better described by jump diffusion rather than simple diffusion as demonstrated by the differences in Schwarz criteria. Furthermore, there is some indication that GARCH(1,1) combined with jump diffusion is marginally more descriptive than jump diffusion alone. Table 3.6 shows detailed

TABLE 3.5 Maximum Log likelihood Values and Schwarz Criteria (SC)

	NP15		Gas		Crude	
	$L(\Theta; y)$	SC	$L(\Theta; y)$	SC	$L(\Theta; y)$	SC
PD	60	-109	919	-1827	992	-1973
JD	148	-267				
D + Arch	84	-151	917	-1817	991	-1965
D + Garch	121	-219	919	-1815	995	-1967
JD + Garch	167	-294			993	-1946

(No value means that the regression did not converge)

TABLE 3.6 Regression Results for NP15 (t-statistics for each parameter are shown in parentheses)

Parameter	Pure Diffusion (PD)	Jump Diffusion (JD)	Jump Diffusion + GARCH(1,1) (JD + Garch)
κ	0.001 ± 0.028 (0.01)	0.012 ± 0.017 (0.67)	-0.001 ± 0.015 (0.04)
σ^2	0.249 ± 0.008 (32.84)	0.066 ± 0.008 (8.77)	
α_0			0.033 ± 0.008 (4.22)
α_1			0.349 ± 0.095 (3.66)
β_1			0.208 ± 0.094 (2.21)
ϕ		0.131 ± 0.033 (3.99)	0.063 ± 0.023 (2.76)
μ		-0.086 ± 0.214 (0.40)	0.209 ± 0.400 (0.52)
γ^2		1.391 ± 0.385 (3.61)	1.538 ± 0.861 (1.79)

regression results for Pure Diffusion, Jump Diffusion, and Jump Diffusion + GARCH(1,1) on NP15 electricity prices.

We see that the models are generally consistent with each other. The mean return κ and mean jump size μ are small and poorly determined in all models. The diffusion variance σ^2 in the JD model is roughly consistent with the unconditional variance of the D + Garch model, calculated to be 0.075.³ Furthermore, the JD total variance, calculated to be 0.248,⁴ is consistent with the PD model's variance of 0.249, while the JD + Garch model total variance of 0.172 is somewhat lower than expected. Interestingly, the jump frequency ϕ in JD is twice that of the D + Garch model, while the jump variance γ^2 is smaller (although the difference is statistically insignificant). It is tempting to suggest that perhaps some of the intermediate valued returns are being accounted for differently in each model. In the JD model, there is little choice but to classify them as jumps. In the D + Garch model however, it may be possible to attribute them to the GARCH process, particularly if they tend to cluster in time.

While comparative tests like the likelihood ratio test and the Schwarz criterion offer an indication of the relative quality of each model, they do not yield an absolute assessment. Many more tests, just a few of which have been touched on this section, are required to determine if the hypothesised specification yields a model that statistically identical to the observed data.

As we alluded to at the beginning of this section, a useful first test is often a simple qualitative inspection of a sample price process and the associated log return distribution.

We conclude this chapter then with figure 3.5, which is divided into panels A, B, and C and show, respectively, comparisons of simulated PD, JD, and JD + Garch prices with the actual NP15 data. In each panel, the left graph shows a simulated price history in grey and the observed NP15 data in black. The panels' right graphs depict the

³ The unconditional variance for a GARCH(1,1) process is $\sigma^2 = \frac{\alpha_0}{(1 - \alpha_1 - \beta_1)}$

⁴ We define the total variance for the jump models to be equal to $\sigma^2 + \phi \cdot \gamma^2$, where σ^2 is the unconditional variance.

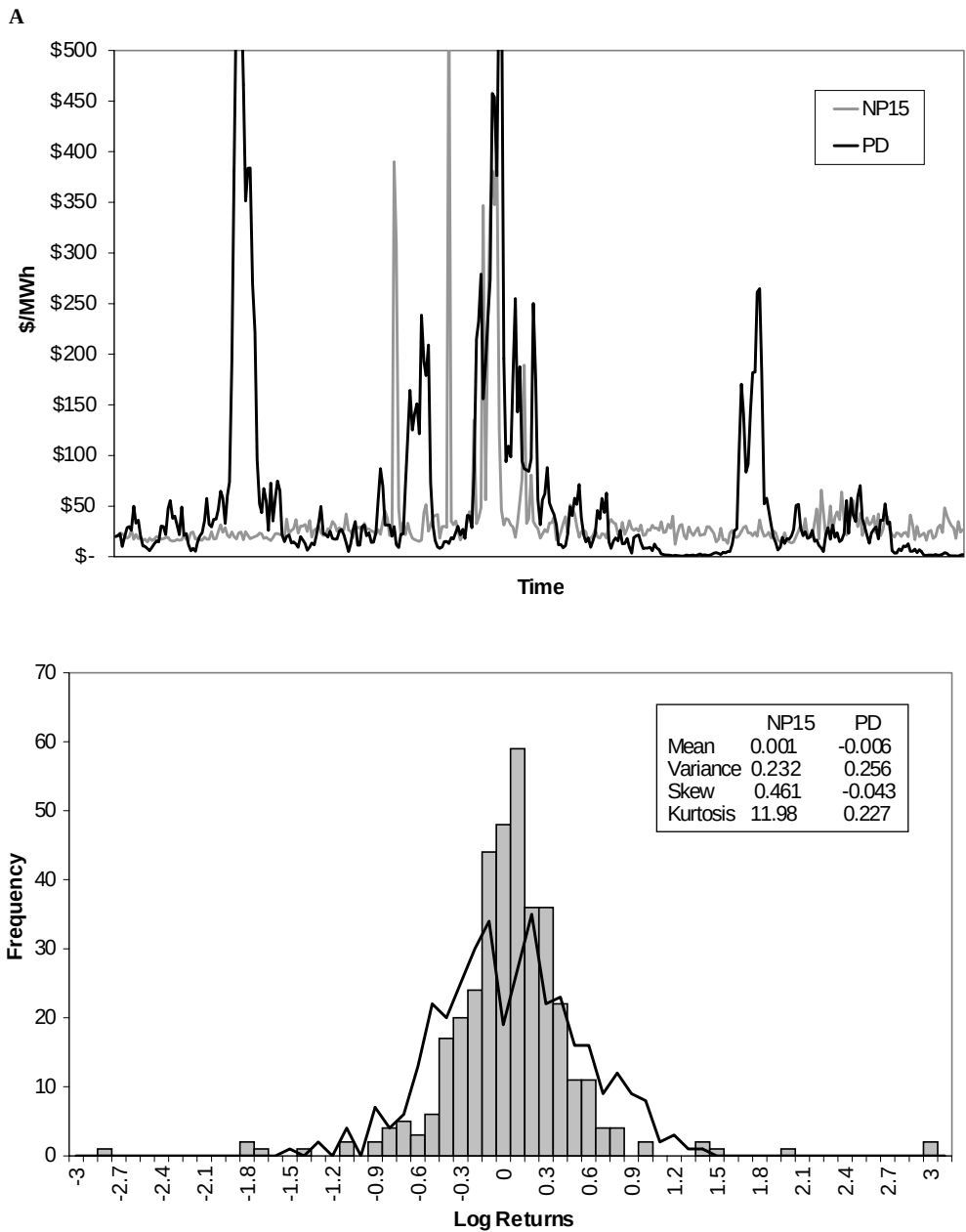


FIGURE 3.5A Comparison of Simulated Price Processes with NP15 Data. A: Pure Diffusion

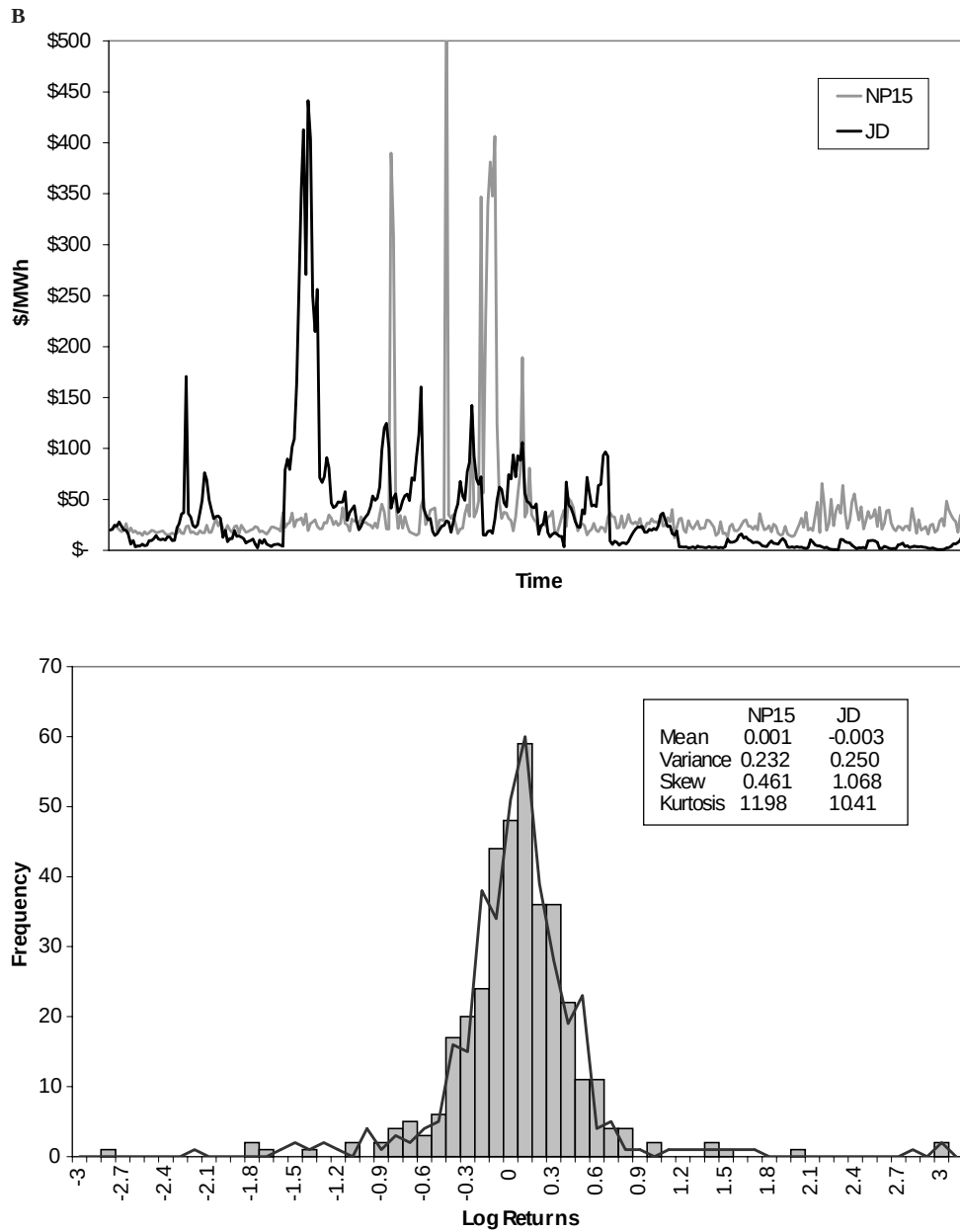


FIGURE 3.5B Comparison of Simulated Price Processes with NP15 Data. **B:** Jump Diffusion

C

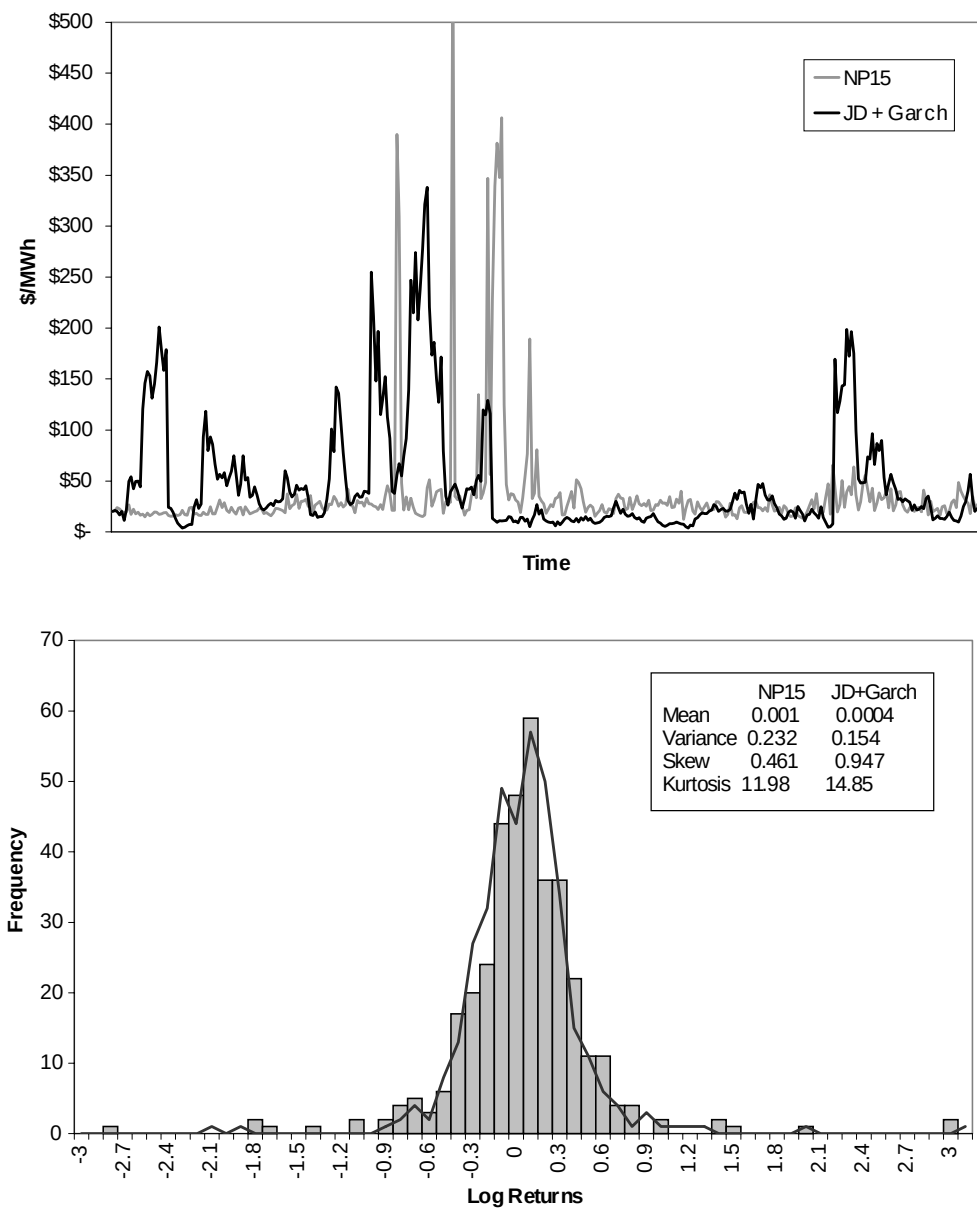


FIGURE 3.5C Comparison of Simulated Price Processes with NP15 Data. C: Jump Diffusion with GARCH(1,1)

distribution of NP15 daily log returns as a histogram and daily log returns for the simulated prices as an overlaid black line. Casual inspection shows that while the Jump Diffusion models do yield more realistic price processes in a number of important respects, they nevertheless produce price paths that are clearly distinguishable from the observed data.

3.5 Summary

In this chapter we have discussed volatility modelling and estimation in the energy commodity markets, emphasising the difference between this market and financial markets. We discuss the estimation of volatility from both historical and implied data, again from the perspective of the energy user. We further discussed a number of stochastic volatility models and have shown how to estimate the models via ordinary least squares and maximum likelihood. We have tested our estimation techniques on a number of examples drawn from energy markets including electricity, gas and crude oil.